DRAFT FOR AUTHOR REVIEW · NON_CITABLE · NO_HUMAN_SIGNOFF_D085 · Contributions and presentation details pending.

CulinaryState — integrated report draft, 11 November 2026

Group 17: Kush Tokas and Parv Goyal. **AI_DRAFTED / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.** This is a reviewable report draft, not a final submission or scientific validation. The authors must supply their own contribution statement and final edits. Presentation date, slot and length remain unconfirmed.

Abstract

CulinaryState implements a recipe-state engine, a process-effect evidence resource and an evidence-aware answer pipeline. Engineering runs cover the recipe corpus, but the scientific claims remain limited by provisional lineage, unverified evidence and non-independent gold. The D5 benchmark contains 74 questions. Both H1 and H3 are RULE_NOT_MET under the recorded decisions; neither result is repaired by a favourable secondary statistic. The AA pilot's frozen fit uses two papers and remains NOT_VALIDATED / RESEARCH_ONLY. The report describes these limits and the negative results alongside the engineering outputs.

Every result below retains the labels and denominator of its source section. AI-delegated approvals under D-085 are not human verification or independent inter-rater evidence. Historical human audit results are identified as such. Original frozen artifacts remain unchanged; this document assembles existing evidence, not a new run.

Research questions, benchmark and scope

Proposal H1 concerns state-aware retrieval; H2 concerns process information versus food identity; H3 concerns unsupported claims with abstention; H4 concerns feasibility and recovery. The SSOT uses different H2–H5 labels. The explicit mapping is in `reports/final/sections/S179_hypothesis_and_task_crosswalk_DRAFT.md`. D-080 accepts the Task 5/6 adaptation merge. It does not imply that every task was evaluated.

D-090 freezes S1=12, S2=12, P1/P2/P3/C=10 each and U/X=5 each (74 total), below the planned S sample. D-089 and `registry/h8_scope_cut_register_v1.csv` declare full H2, H4 and later learning lanes not run before the talk. Step 35 is designed, not run. These omissions remain part of the scope disclosure.

H1 result and provenance

NON_CITABLE / MODEL_ASSISTED_GOLD / CIRCULAR_SELF_CONSISTENCY_ONLY / NO_HUMAN_SIGNOFF_D085. H1 uses 24 AI-gold recipe questions (22 Gutenberg and 2 Wikibooks). Near-duplicate leakage is not controlled; no source-family-disjoint result or independent human gold is claimed. Accuracy is B0=2/24 and B1=B2=B4=6/24. B4 versus each comparator has 4 discordant pairs and exact two-sided McNemar p=1.0. D-092 records **RULE_NOT_MET**, not equivalence between methods and not evidence of absence of an effect. The small sample cannot sustain a precise comparative conclusion.

The original `reports/del5/d5_score_v1_D085/d5_scoring_result.json` leaves H1 NOT_COMPUTED due to unresolved answers. The separately retained `d5_h1_after_rung_blind_check_D085.json` in that directory resolves six answers, all as wrong, and supplies the H1 result cited by D-092. This supplement is AI-delegated review, not human adjudication. The report preserves the distinction rather than overwriting the original scoring artifact.

D1: evidence resource and its limits (S179 report draft, 2026-11-11)

Status: AI_DRAFTED. All counts below describe imported records: E3_UNVERIFIED, COND_UNAUDITED, BOOTSTRAP_UNVERIFIED, NON_CITABLE. This is a resource description, not a performance or evidence-quality claim. New report assembly is AI_PRECHECKED; no human review of this new section is recorded.

Sources, targets and tiers

The evidence resource contains **17,978 ACTIVE rows**, with **178 deprecated ids excluded**. Its sources are Bognár (7,090 rows), USDA retention factors R6 (7,018), and Phenol-Explorer 3.6 (3,870). Every ACTIVE row is E3; 14,108 are UNVERIFIED and 3,870 AI_PRECHECKED. **Human-verified D1 rows: 0/40.** These are counts of database imports, not independent primary experiments. Source: `reports/s178/analysis/evd_dataset_card_v1.json`, `s203_9_1_total_rows`, `s203_9_6_evidence_tiers` and `labels`.

There are 224 SVAR target ids (213 canonical variables after duplicate mapping). By target subfamily, the row counts are NUTRIENT_CONTENT 12,116; PHENOLIC_COMPOUND_CONTENT 3,870; GROUP_AGGREGATE 1,496; MASS_YIELD 496. These categories sum to the ACTIVE total. Source: same JSON, `s203_9_5_target_types`.

Conditions: preserve missingness

Every figure in this table is an E3_UNVERIFIED / COND_UNAUDITED diagnostic count. No UNKNOWN is filled with zero.

Axis	MEASURED	Two-sided RANGE	One-sided censored	UNKNOWN	Not recorded	NOT_APPLICABLE
Temperature	2,430	216	3,369	11,963	0	0
Duration	2,607	476	0	14,895	0	0
pH	0	0	0	17,978	0	0
Moisture	0	0	0	17,978	0	0
Pressure	129	0	0	0	10,831	7,018

Source: same JSON, `s203_9_9_condition_coverage.<axis>.rows_by_status`. Water activity is in the schema but never populated (0/17,978). Temperature and duration are both stated on 1,575 rows (8.76%), all from Phenol-Explorer. A stated condition does not imply it has been checked against the primary paper.

Supply against recipe demand

The paired coverage analysis is distinct from condition completeness. EVD has 338 routable (ING, PROC) pairs. A seeded 1,000-recipe non-BBC sample has 12,501 distinct pairs, of which 111 (0.89%) have an EVD pair. At process level, 63.47% of 4,724,493 non-BBC process events use a process with some EVD row; this is an upper bound that ignores the ingredient. For vitamin C, 102 of 125,172 full-corpus distinct pairs have an EVD pair (0.08%). Source: `reports/s178/evd/evd_coverage_vs_corpus_demand_v2.json` and its companion Markdown table. These are different denominators and must not be collapsed into one coverage percentage.

Figure for report/deck: `reports/s178/evd/fig_evd_coverage_by_family_v2.svg`, whose adjacent JSON/CSV carry the numbers. Its caption must retain NON_CITABLE / E3_UNVERIFIED / COND_UNAUDITED and distinguish full-corpus process counts from sampled pair counts. Tables above are the source/target and condition panels; no new scientific run was performed for this section.

Lineage and permitted interpretation

There are 127 provisional independence groups: Bognár 1, USDA 1 and Phenol-Explorer 125. The feasibility report's headline under reading A is **publication-component CV**, not confirmed study-disjoint evaluation. Reading B, which also requires human verification, has zero eligible rows and uses leave-one-source-family-out. Source: dataset-card JSON `s203_9_7_study_source_counts` and `exports/lineage/step08a_v1/feasibility.json` / `feasibility_report.txt`. The human edge audit G08-3 is FAIL; exact audit counts and slide wording are in `S179_H2_and_lineage_statements_DRAFT.md`. Imported row count cannot be read as independent study count.

No claim is made about comprehensive chemistry coverage, primary-paper correctness, study-independent prediction, or human-verified evidence. Phenol-Explorer raw-table redistribution remains restricted; only counts and ids enter this section. See `S179_licence_and_split_labels_DRAFT.md` and D-079.

Acceptance remaining: human review of this new section/figure captions and incorporation into the final report and slides. D-079 accepted earlier drafts; it is not a review of this newly written section.

Source section: `reports/final/sections/S179_D1_evidence_resource_DRAFT.md`.

Report section draft: D3 state-tracking engine (S179)

Status: **AI_DRAFTED** for the final report (board items T9-C16-R1 and T3-C17-R2). Counts only. No figure here is an accuracy claim. Written 2026-11-10 by the Mac agent on `kush/s179-finish-fixes`.

One-line verdict for the report. D3 is built and has run on the full corpus. Its replay is deterministic and catches every planted structural error. Gold comparison is limited to 7 P1 recipes and 40 AI-assisted development recipes, none of them independent; on those, the rule compiler ends with the gold final state on 11-20 % of state fields.

1. What D3 is and where it ran

- **Pipeline.** Recipe text goes through the rule compiler (`state_graph.rule_compiler`, also Rung A's verb slot) to an event log. The replay engine (`state_graph.engine`) turns the log into a per-step state graph.

- **Corpus run.** The Step 18 replay ran on every non-BBC recipe with instructions: 599,971 texts (Food.com 522,517; Gutenberg segments 73,778; Wikibooks 3,676).
- Source: `reports/s178/corpus/replay_invariants_corpus_v2.json`, sha256 `2abe7ae7…`, run RUN-S179-CORP-002.
- Not run: TASTEset has no instruction text, and BBC is not on the server (D-003).

2. Metrics that need no gold (T9-C16-R1: replay consistency and property tests)

Check	Result	Source
Same-worker determinism (texts replayed twice)	12,046 of 12,046 identical (Food.com 10,416; Gutenberg 1,550; Wikibooks 80)	replay invariants v2
Cross-process determinism (PYTHONHASHSEED 0 vs 1)	12,046 of 12,046 identical snapshot and JSONL	same
Engine vs an independent re-derivation of the 7 structural invariants	0 disagreements on 599,971 texts	same
Planted errors detected at the planted position: double consume	28,854 / 28,854	same (mutation power)
Planted errors: order swap	29,945 / 29,945	same
Planted errors: reuse after discard	902 / 902	same
Planted errors: split fractions above one	390 / 390	same
Texts whose replay FAILs an invariant	12,774 of 599,971 (2.1 %)	same

Limits. - These tests show the engine applies its own rules consistently. They say nothing about whether the events match the recipe. - The split-sum check is almost empty: 0 of 88,817 partition groups have a checkable basis and fraction. - Lot accounting, invented values and quantity propagation are not checked by replay (`not_checked_by_replay`).

3. Gold comparison (T3-C11-R1, T3-C17-R2)

Sources: - `reports/del3/compiler_vs_gold_pooled_S179.json` (RUN-S179-GOLD-001) and `reports/del3/event_type_table_S179.json` (RUN-S179-GOLD-002); - method and caveats: `reports/del3/compiler_vs_gold_pooled_S179.md`.

All three sets are **NOT INDEPENDENT**. The engine was developed with them in view, and every set began as an AI draft. Split label: near-duplicate leakage not controlled.

Pooled counts, Wilson 95 % intervals:

Set	n	Label basis	Final state fields correct	Loose-event F1	Event-type F1
P1, side A	7	HUMAN_ACCEPTED_AI_DRAFTS_NOT_BLIND	5 / 31 = 0.16 [0.07, 0.33]	0.24	0.61
pilot_20 v1.1 (BBC)	20	AI-assisted, human-reviewed dev gold	9 / 79 = 0.11 [0.06, 0.20]	0.27	0.69
Food.com v1.1	20	AI-assisted, human-reviewed dev gold	24 / 120 = 0.20 [0.14, 0.28]	0.32	0.71

Event types at level P, P1 set (7 recipes). Matched is the per-recipe count match, min(gold, predicted). Intervals come from a recipe-cluster bootstrap with 2,000 resamples.

Type	Gold	Predicted	Matched	Precision	Recall
Merge	36	21	21	1.00 [1.00, 1.00]	0.58 [0.43, 0.72]
Transform	35	63	33	0.52 [0.40, 0.70]	0.94 [0.81, 1.00]
Transfer	22	6	6	1.00 [1.00, 1.00]	0.27 [0.05, 0.47]
Split	8	1	0	0.00	0.00
Discard	1	1	0	0.00	0.00
Reserve	0	3	0	0.00	n/a

The dev sets show the same shape: - Transform: precision 0.54-0.55, recall 0.96-0.98. - Merge: precision 0.97, recall 0.67-0.74. - Transfer and Split: under 0.6 on recall in both sets. - Reserve: too few gold events to read (8 and 1).

The compiler emits a Transform for almost every heat or process verb. It misses many merges and most transfers and splits. The P1 handbook finding agrees: two human-accepted passes disagree most on final_state (0.82 per pair) and event_type (0.76).

4. Allowed and forbidden wording

- **Allowed:** "built and run on 599,971 texts"; "replay deterministic and catches 100 % of planted structural errors"; "against 7 P1 and 40 AI-assisted development recipes (not independent), 11-20 % of final state fields match".
- **Forbidden:** "accurate", "validated", "state tracking works", or any number without its n, its label basis and "not independent".
- **D3 accuracy** stays NOT MEASURED until an independent gold exists. That is P1b or later, and the decision on the minimum gold size is human (Q28).

5. Not done

- **Rung A extractor:** the 17a EventGraph is not scored directly; there is no adapter to engine events.
- **P1 side B:** not in this tree.
- **BBC replay invariants:** a Mac run is possible but has not been done.

Source section: `reports/final/sections/S179_D3_state_tracking_DRAFT.md`.

H2 and lineage: report/deck wording (S179 draft, 2026-11-11)

Status: AI_DRAFTED / AI_PRECHECKED. The passages below are ready for report/deck assembly. They do not record a human reading of this new draft. T2-C06-R2 and T6-C04-R3 human-review acceptance remains open.

Lineage statement (T2-C06-R2)

The G08-3 edge-audit gate **FAILED for both reviewers**. In the committed audit, STRONG edges are 50/50 correct for each reviewer, Wilson lower bound 0.9286523998213857. For B2, the audit records 49/50 for Kush (lower bound 0.8950455630610349) and 0/50 for Parv (lower bound 0.0). For B3 the corresponding counts are 13/50 (lower bound 0.15871527430089563) and 0/50 (lower bound 0.0). The gate's Wilson lower-bound threshold is 0.9. Uncertain edges count as errors under the registered conservative rule. These are historical human audit results, transcribed without assigning a new AI reviewer a human identity.

Source, exact fields: `reports/step08/step08a_audit_result_v2.json`, `G08-3`, `edge_precision.<reviewer>.<stratum>.correct`, `.n` and `.wilson_lower`. The miss-rate check is REPORT_ONLY: 2/200 duplicates for Kush and 0/200 for Parv (`THR-08A-18_miss_rate`). It does not repair the edge gate.

The provisional feasibility export has 127 groups, with USDA and Bognár one group each. Its headline under reading A (tier eligibility alone) is **publication-component CV**; the weakest label across target families and readings is **leave-one-source-family-out**. Under reading B (also human verification), eligible pool n=0. Phenol-Explorer has 125 publication-key groups; these are not confirmed experiment boundaries. Lab-disjoint feasibility is NEED_DATA. Source: `exports/lineage/step08a_v1/feasibility.json` and `feasibility_report.txt`.

One-line implication for every H2 figure: provisional publication keys and the failed lineage edge audit prevent reading these diagnostics as evidence of generalisation across independently verified studies. USDA and Bognár have no study-disjoint estimate.

Honest H2 slide text (T6-C04-R3)

Title: Process information versus food identity: diagnostic, no direction established.

Labels applying to every slide number: DIAGNOSTIC / BOOTSTRAP_UNVERIFIED / NON_CITABLE / PRIOR_EXPOSURE_REANALYSIS / PROVISIONAL_08A / COND_UNAUDITED / NO_HUMAN_SIGNOFF_D085.

On Phenol-Explorer's provisional publication/study-key-disjoint split, process-mean loss minus ingredient-mean loss is **0.0992 [−0.0446, 0.2401]**, with **362 rows / 20 clusters**. Positive means lower ingredient-mean loss; the interval crosses zero. Matrix-disjoint favours the process arm by construction; process-disjoint favours the ingredient arm. Only the provisional study-key split compares the arms on equal terms.

Source: `reports/step29/diag/diag_results_S179.json`, `result.families.PHENOL-EXPLORER-3.6.contrast_d2_vs_b2.splits.study_disjoint`.

Bognár repeated-value share is **0.9489** and USDA **0.5874** within their diagnostic subsets (`duplicate_control.share_of_rows`); each source is one lineage group. The original data were already used by DEL-2 and the exploratory Step 30 run. This is prior-exposure reanalysis, not a fresh independent test. D7's Phenol-Explorer random predictions equal the global mean D1 exactly (`d7_vs_d1_max_abs_pred_diff = 0.0`); no learned-model advantage follows.

Conclusion to show: Full proposal H2 is DECLARED_NOT_RUN before the talk (D-089; H8-02). The diagnostic cannot replace that test. No process-information advantage or independent-study generalisation is claimed.

Review and assembly boundary

The writer checked these statements against the cited committed files. This is an AI precheck, not a human signature or inter-rater result. The original audit, feasibility export, failed gate and frozen outputs remain unchanged. Human review required by the board and incorporation into the final deck remain pending.

Source section: `reports/final/sections/S179_H2_and_lineage_statements_DRAFT.md`.

D4: evidence-aware retrieval and abstention (S179 report draft, 2026-11-11)

Status: AI_DRAFTED · DIAGNOSTIC · BOOTSTRAP_UNVERIFIED · NON_CITABLE · NO_HUMAN_SIGNOFF_D085. D5 uses MODEL_ASSISTED_GOLD and author-only review (AI-delegated). This section transcribes committed outputs; no scorer, benchmark, prompt or frozen result is changed.

Frozen-benchmark scope and the headline result

D-090 froze n=74 questions: S1 12, S2 12, P1/P2/P3/C 10 each, U/X 5 each. D4's answer/risk analysis uses the 40 P/C questions. Rungs ran in RUN-S179-D5-001. Freeze `D5-BENCH-2026-11-10b` added the EVD and relatives scorer inputs after rung execution started and before scoring (D-092); benchmark/scorer/prereg hashes were unchanged. No independent human gold or inter-rater result exists. Split label: near-duplicate leakage not controlled; this is not a source-held-out benchmark.

H3 is **RULE_NOT_MET** (D-092). The exact two-sided sign test has 16 discordant questions in B7's direction, 0 in the other direction, p=0.000030517578125. The additional P3 condition fails: B7 abstains 0/10 against B2 3/10. Therefore the sign-test result alone cannot establish the preregistered conjunction. The 30-claim veto audit was NOT_RUN because it cannot turn an already failed conjunction into a pass. B7 is **rule-based evidence-sufficiency abstention (uncalibrated)**. NEG-0051 records the mismatch between the gold's exact/parent food evidence boundary and the decision layer's broad cross-food evidence boundary.

Source for the test: `reports/del5/d5_score_v1_D085/d5_scoring_result.json`, `h3_selected` / `ood_abstention`, plus D-092. The raw Markdown rounds the displayed p to 0.0; the exact JSON value above is retained here.

Retrieval (M1)

Source throughout: same score JSON, `retrieval_d4_m1.rungs.B2.blocks.pooled_P1_P2_C`. B2 has retrieval lists for **20** questions; the pooled label does not mean all 30 P1/P2/C questions were scored. Ten questions lack a list (`n_questions_without_list`). Rungs B0/B5/B5T/B7 expose no retrieval list and are NA_NO_RETRIEVAL_LIST, not zero-recall systems.

K	Plain recall (mean per question)	Compatibility recall (as recorded)
1	0.2	0.0
3	0.3417	0.0
5	0.375	0.0
10	0.375	0.0

MRR is 0.3417, explicitly EXPLORATORY_NOT_IN_PREREG. No question states a condition axis (n=0). The nested JSON note claims compatibility recall equals plain recall by construction, but the recorded compatibility vector is 0.0 at every K while plain recall is nonzero. This internal output inconsistency is retained and flagged; no compatibility-recall claim is made from it, and the frozen output is not silently repaired.

The `source_disjoint_diagnostic` field stratifies by **gold source family**: USDA 16 questions, Phenol-Explorer 3, Bognár 1, labelled DIAGNOSTIC_ONLY. It is not a retraining or retrieval run with a source family held out. P3 returned evidence on 10/10 questions; that rate is descriptive and is not recall.

Answer risk and coverage (M2)

Source: same JSON, `d4_m2_risk_coverage.decision_layer_rungs.<rung>.primary_precision_class.points`. Risk is the recorded E1 error count divided by the included question count, and thresholds are ordinal rules, not calibrated probabilities. Every cell below is a NON_CITABLE diagnostic on the 40 P/C questions.

Inclusion threshold	Coverage	B5 errors/n	B5T errors/n	B7 errors/n
QUANTITATIVE	0.45	12/18	0/18	2/18
INTERVAL	0.75	18/30	2/30	4/30
QUALITATIVE	0.975	27/39	5/39	13/39

Inclusion threshold	Coverage	B5 errors/n	B5T errors/n	B7 errors/n
ABSTAIN	1.0	28/40	5/40	14/40

The ABSTAIN threshold is the endpoint that includes all questions; coverage 1.0 is not an assertion that all questions received substantive answers. C-block agreement is 9/10 for B5, B5T and B7; each reports cannot-separate on 10/10. This scorer field is descriptive, not a separate hypothesis result (`c_block.rungs`). Router accuracy is only 6/10, status INCOMPLETE_GOLD_BRANCH, with X controls 4/5 (`router`); it must not be presented as accuracy over all 74 questions.

Earlier retrieval development study (silver; not D5)

`reports/del4/retrieval_silver_v2_analysis.md` analyses 600 templated development queries over 17,978 EVD rows. These are **SILVER_DEV_ONLY / NON_CITABLE** figures, not independently labelled questions or recipe-step chunks. RRF BM25+e5-large-v2 MRR@10 was 0.762 versus 0.704 for the bge-small RRF list, paired delta +0.057 [0.035, 0.082]. Structured+dense R4 versus structured R3 had delta +0.005 [-0.006, 0.015]; the dense blend was not promoted. The source analysis notes oracle-slot R3=0.856. Model selection used this dev study, so it is not an independent test of the selected embedding. The frozen D5 run, with its different task and small denominators, is reported separately above.

Limits and acceptance remaining

The gold is AI-drafted/AI-reviewed, not independent human gold. Qualitative answers found by broad cross-food retrieval do not resolve the P3 definition mismatch. The formal H3 rule remains failed; no post-hoc boundary choice replaces D-092. The raw score's H1 NOT_COMPUTED is also retained: the separate rung-blind supplemental check gives H1 RULE_NOT_MET (see the hypothesis crosswalk).

This report text satisfies the evidence-traceability portion of T8-C19-R1. Final report/deck assembly, human reading where required, and an amendment-only resolution of the recorded compatibility-recall inconsistency remain separate work. No claim of calibration or scientific validation follows from these diagnostics.

Source section: `reports/final/sections/S179_D4_retrieval_and_abstention_DRAFT.md`.

AA pilot: frozen Model A identifiability diagnostic

Date: 2026-11-11. Board item: T5-C12-R2.

DIAGNOSTIC / NON_CITABLE / AI_PRECHECKED / NO_HUMAN_SIGNOFF_D085. Model A remains **NOT_VALIDATED / RESEARCH_ONLY**. Reviewer: author-only review (AI-delegated). This report profiles the already frozen estimates; it neither refits Model A nor opens SIM-VAL. No parameter-precision gate is passed by this report.

Inputs and reproducibility

- Frozen fit: `reports/step10/aa_pilot_model_a_fit_D085.json`, SHA-256 `b9247db03e98423f8866fa50b24e2ecd33831d907f3aace00b1f7e4f2ea9543f` (D-091).
- Frozen model: unweighted OLS on ln k, separate paper intercepts and shared Ea, `src/culinarystate/simulation/aa_pilot_fit.py`, version `aa-pilot-fit-0.1.0-draft`.
- New tool: `tools/s179_aa_profile_summary.py`; new output: `reports/step10/aa_identifiability_profile_summary_S179.json`. The tool checks both code hashes recorded by the frozen fit, refuses an existing output path and records the fit/tool SHA-256 values.
- Command: `python tools/s179_aa_profile_summary.py --fit reports/step10/aa_pilot_model_a_fit_D085.json --out reports/step10/aa_identifiability_profile_summary_S179.json`.
- Source rows are AI_PRECHECKED under D-085, not human verified. There are **7 included rows from 2 papers, 3 regression coefficients and 4 residual degrees of freedom**. This diagnostic used frozen sufficient summaries before the private CAL transfer arrived. The later source review and Model B work are reported separately below; the profile calculation is unchanged.

All numerical results below have the four diagnostic labels above. No bootstrap was performed.

Method and why the summaries are sufficient

Let beta be the three regression coefficients, C the frozen OLS covariance, s the frozen residual SD, n = 7 and nu = 4. The frozen fitter uses C = s²(X'X)^-1 and SSR_min = nu s². For coefficient j fixed at b, minimising over all nuisance coefficients gives the exact quadratic profile

$$SSR_{profile}(b) = SSR_{min} + s^2 (b - \beta_j)^2 / C_{jj}.$$

Profiling the unknown common Gaussian variance gives

$$LR(b) = n \log[SSR_{profile}(b) / SSR_{min}].$$

This algebra reconstructs the unconstrained profile from the committed summary without raw observations, refitting, a scan-width choice or a new optimiser. The 95% **asymptotic** likelihood interval uses `LR <= chi²_1(.95) = 3.841458820694124`; its half-width is `sqrt(C_jj * nu * expm1(chi²_1(.95) / n))`. The confidence level is a disclosed diagnostic choice, not a new acceptance threshold.

Because only four residual degrees of freedom remain, the table separately includes the Student-t interval `beta_j +/- t_4(.975) sqrt(C_jj)`. This is finite-sample conditional inference under the fixed-design, independent, homoscedastic Gaussian-error model; it does not account for paper dependence, verification error, assay mismatch, model misspecification or extrapolation to another matrix. The t interval is wider here. The likelihood profile is exact algebraically; its chi-square confidence calibration is asymptotic.

Method references: the NIST likelihood-ratio test description gives the chi-square calibration, while NIST's linear-regression uncertainty example uses residual degrees of freedom for Student-t inference. The displayed profile identity is derived above from the frozen fitter's OLS equations, rather than inferred from an unscaled covariance condition number.

Per-parameter results

Reference temperature is 363.15 K. The ln k_ref intercepts use k in min^-1 as in the frozen AA fitter. Ea is displayed in kJ/mol; the JSON retains J/mol.

Parameter	Estimate	Asymptotic profile 95% interval	Conditional Student-t 95% interval	Profile status	Practical gate status
ln k_ref, AAK-01	-4.3152	[-4.5445, -4.0859]	[-4.6875, -3.9429]	BOUNDED_CONDITIONAL_ON_OLS_MODEL	NOT_ASSESSED_NO_LOCKED_PRECISION_CRITERION
ln k_ref, AAK-06	-3.1251	[-3.4142, -2.8360]	[-3.5944, -2.6558]	BOUNDED_CONDITIONAL_ON_OLS_MODEL	NOT_ASSESSED_NO_LOCKED_PRECISION_CRITERION
Shared Ea, kJ/mol	64.3978	[46.6419, 82.1536]	[35.5707, 93.2249]	BOUNDED_CONDITIONAL_ON_OLS_MODEL	NOT_ASSESSED_NO_LOCKED_PRECISION_CRITERION

The recorded covariance is positive definite; within the fitted linear model there is no flat coefficient profile. This establishes bounded model-conditional intervals, not sufficient scientific precision for a use case. No locked precision or relative-width criterion was found in the AA pilot preregistration. Proposed G26-CAL settings must not silently become signed rules. A raw covariance condition number would mix units of dimensionless ln k and J/mol; no such number is presented as an identifiability test.

Checks and limits

`tests/unit/test_s179_aa_profile_summary.py`: **10 passed** under Python 3.11.9. A failing collection first confirmed the new tool was absent. Synthetic constrained OLS refits at both endpoints for every coefficient reproduce the likelihood-ratio cutoff. Tests distinguish profile and t intervals, preserve the caller's input, and refuse invalid degrees of freedom, residual SD, covariance or paper identities.

The new tool reads no recipe text, CAL sheets or SIM-VAL material. It checks existing hashes rather than replacing the fit or its code. Existing failed gates and frozen outputs remain unchanged.

- Two papers provide very little information about between-paper heterogeneity. The fitted between-paper SD is not a random-effects variance estimate with a profiled confidence interval here.
- Residual checks, leverage and source metadata cannot be reconstructed from summaries alone. The later CAL transfer enabled source/unit prechecks and per-paper Model B; see the dated AA update below.
- The likelihood profile assumes a common Gaussian residual variance and no within-paper dependence. Bounded intervals under that model are not evidence of robust practical identifiability under alternatives.
- The report supplies parameter profiles and explicit parameter statuses for T5-C12-R2. Its practical gate remains unassessed; a reviewer may accept the report artifact without approving a scientific gate.
- PARAM minting, SIMULATE calibration, domain/basis checks, SIM-VAL extraction and prediction, and the T5 G15 scorecard retain their own requirements. This report changes none of them.

Source section: `reports/step10/identifiability_report_v1.md`.

Report section draft: data licences and split labels (S179)

Status: **AI_DRAFTED** for the final report and the release card (board items T9-C28-R1 and T9-C09-R3). D-079 records Kush's acceptance of the licence paragraph. It states what the registries record and does not verify or change any licence. Written 2026-11-10 by the Mac agent on `kush/s179-finish-fixes`.

Sources: `registry/recipe_source_usage_registry_v1.csv`, `registry/data_source_acquisition_manifest.csv`, the SRC ledger, `DECISIONS.md` D-003, `src/culinarystate/decision/public_mode.py`, `registry/split_registry_v1.csv` and `reports/s178/leakage_audit_d5_findings.md`.

1. Licence and redistribution paragraph (T9-C28-R1)

> **Data, licences and what is shown.** The project reads five recipe sources. Their status, as recorded in `registry/recipe_source_usage_registry_v1.csv`, is: - **BBC Good Food** (11,934 recipes, scraped for an earlier course assignment) is third-party copyright, LOCAL_RESEARCH_ONLY. Its text stays on one researcher's machine. It never goes to a server or an external service, and it is never committed (D-003). - **Food.com** (Kaggle download) is declared CC0 by the uploader, but that authority is not verified (LICENCE_UNVERIFIED), so its text is not redistributed either. - **TASTEset** is LICENCE_UNVERIFIED in the usage registry. The acquisition manifest notes an MIT licence, but no source row confirms it. - **Wikibooks Cookbook** pages are CC BY-SA 4.0, with attribution by page and revision. - **Project Gutenberg** cookbooks (338 books) are public domain in the USA under the Project Gutenberg licence.
>
> Process-effect evidence comes from: - the USDA retention tables (US government work, public domain); - Phenol-Explorer 3.6, a free public resource whose redistribution needs the authors' permission. We therefore publish only derived counts and identifiers from it, never its tables. The publications it cites remain under publisher copyright and are cited, not reproduced.
>
> Everything committed or shown in this report, the slides and the demo screenshots is counts, identifiers, hashes and figures computed from them. No recipe text from BBC Good Food or Food.com appears.
>
> The demo has a public mode for anything that is not the researcher's own machine. It drops every knowledge-graph edge computed over the BBC corpus before any answer is built, and it omits the carbon figures, whose upstream licence is unverified. In local mode, any request that arrives through a tunnel or proxy is refused (HTTP 503).

Update: D-079 accepted the licence paragraph. No new source licence is verified by this assembly.

Source section: `reports/final/sections/S179_licence_and_split_labels_DRAFT.md`.

Limitations and unfinished work

07.A v4 FAIL and G08-3 FAIL remain on record. 07.A-R1 is NOT_RUN under D-085 because two personas of one model cannot supply independent labels. AI_DRAFTED_GOLD does not become HUMAN_BLIND through delegation. The H3 30-claim audit was not run; no positive claim is made in its absence. The compatible-recall inconsistency in D4 remains unresolved in the frozen output. A proposed future evidence-boundary amendment is separate at `reports/del5/D5_P3_evidence_boundary_amendment_PROPOSAL_2026-11-11.md`; it is not adopted or executed.

AA Model B and separate-context SIM-VAL extraction/source precheck are now complete. Source precheck leaves zero computationally eligible held-out rows, so prediction and retention-error assessment remain NOT_RUN. PARAM provenance and calibration/domain integration remain incomplete; synthetic values do not replace missing scientific observations. No clinical, dietary or food-safety recommendation is made by the research demo.

The final 1,000-session emission-contract corpus scan, completed report/deck recipe-text leakage comparison, updated frozen split registry, post-19-Nov MPU score and actual timed rehearsal on 24 Nov remain separate acceptance items. A source-field scan alone would not certify absence of verbatim recipe text.

AA update after the private transfer — 11 November 2026

DIAGNOSTIC / NON_CITABLE / AI_PRECHECKED / NO_HUMAN_SIGNOFF_D085. The restored CAL sheet matches the frozen fit hash. Separate Model B fits and every predeclared sensitivity arm's disposition are now recorded. Removing the non-AA-specific paper leaves one paper and returns NEED_DATA. The licence-rejected all-rows arm remains NOT_RUN. Unchanged-row arms do not count as new robustness results. The pooled primary fit retains its HPLC-versus-iodine-titration assay-basis caveat.

Separate-context SIM-VAL extraction finalized 19 rows; a second estimate-unexposed source precheck marked 6 PASS, 12 FLAG and 1 REJECT. PASS means source fidelity only; computationally eligible rows are 0. The three rate rows remain excluded pending assay-basis review, and no raw time/retention observations were available. AAK-12's licence could not be confirmed, so its text remained unopened. No primary prediction/error metric was emitted.

The guarded tool retains null metrics for zero eligible rows. No PARAM/VAL provenance or validated domain has been supplied; production SIMULATE remains unavailable. The 13-gate T5 scorecard has zero PASS. No scientific gate, human verification or independent agreement follows from having completed these report artifacts.

Sources: `reports/step10/aa_model_b_sensitivities_2026-11-11.md`, `SIMVAL_extraction_precheck_2026-11-11.md`, `SIMVAL_source_precheck_2026-11-11.md`, `calibration_disposition_2026-11-11.md` and `G15_T5_scorecard_2026-11-11.md` in `reports/step10/`.

Author contributions

PENDING — to be written by Kush and Parv from their own contribution facts. No effort allocation or individual scientific contribution is inferred from commit authorship, delegated AI personas or this session.

Reproducibility and review index

- Original run records: `registry/compute_runs_v1.csv`; D5 RUN-S179-D5-001 and 29-diag RUN-S179-DIAG29-001.
- D5 freezes: `registry/freezes/D5-BENCH-2026-11-10.json` and supplementary `D5-BENCH-2026-11-10b.json`.
- Gate provenance: `reports/final/sections/S179_gate_provenance_DRAFT.md`; decisions D-079..D-092.
- Negatives: `registry/negative_results_register_v1.csv`, including NEG-0050 and NEG-0051.
- Acceptance checklist: `reports/orchestration/S179_round6_acceptance_audit.csv` and companion Markdown.
- Artifact versions/limits: `reports/final/RELEASE_CARD.md`.
- Proposed work windows: `reports/s179/schedule_to_talk_DRAFT_2026-11-11.md`.

Frozen outputs must not be edited, retuned or silently rescored. Any new experiment requires its own dated amendment, freeze, output path and run record, with the original result retained beside it.

Engineering provenance audit update — 12 November 2026

ENGINEERING_DIAGNOSTIC / NON_CITABLE / NO_HUMAN_SIGNOFF_D085. A separate lexical/structural scan covers all 346 saved D5 answers and 1,000 metadata-template public-demo calls. Thirty saved records have numeric-reference or precision-cap flags (B0:4, B2:8, B5:18); B5T/B7 have no flags within the instrument's scope. The corrected scan finds no demo flags, but only 173 calls reach TRANSFORMATION and the sample does not represent real user queries. Two hundred synthetic recipe requests yield 36 recipe payloads and 51 physical steps: test inputs, not measurements.

The original scan is retained beside corrected v2. Matching a number does not establish food/process/target/ condition applicability. Units, semantic/option entailment, full per-number basis and branch coverage still need work. T0-C06-R2 remains PARTIAL and NEG-0052 retains the engineering finding. No H1/H3 score or frozen answer changed. Source: `reports/s179/emission_audit_2026-11-12.md` and its referenced JSONs/tests/path inventory.

Local completion supplement — 12 November 2026

ENGINEERING_DIAGNOSTIC / NON_CITABLE / NO_HUMAN_SIGNOFF_D085. This section supersedes earlier missing-input and next-action descriptions for the listed work only. Original scientific results and failed gates remain unchanged.

Artifact / split	Local result	Evidence
Required tier-1 bundle	Original archive checksum matches; identity features, Food.com and Gutenberg restored; existing content not overwritten	`reports/s179/local_tier1_restoration_2026-11-12.json`
macOS archive sidecars	Ignored AppleDouble/AppleSingle files quarantined outside checkout; source annotation content preserved	`reports/s179/local_metadata_quarantine_2026-11-12.json`
Actual frozen D5, 74 questions / eight blocks	New v3 registry and complete exact-line exposure section; overlap disclosed, independence unestablished	`reports/s179/d5_frozen_split_leakage_v3.md` and JSON
P/C gold resource exposure	All 79 distinct gold EVD IDs already occur in DEL-2; shared C/P1=2, C/P2=1, P1/P2=9	Same v3 audit
S NER surface exposure, nine ingredient lines from two Wikibooks recipes	RecipeDB3 train=1/9, test=1/9, calibration/dev=0/9; TASTEset any split=2/9; 22 Gutenberg segments have no ingredient-line channel	Same v3 audit; not a NER accuracy test
Strict-v2 public-filtered local demo, 1,000 metadata calls / 200 synthetic recipe requests	155 ABSTAIN, 186 ABSTAIN_SAFETY, 659 ASK; zero table retention figures; 53 physics steps, 141 numeric fields, zero omitted or UNKNOWN numeric provenance in enumerated display fields	`reports/s179/strict_runtime_audit_2026-11-12.json`
Loopback HTTP, public-filtered and local modes	Health, HTML and three requests per mode pass; local forwarding header refused with 503	`reports/s179/strict_http_smoke_2026-11-12.json`
Required learned NER checkpoint	Not available in inspected local inputs; optional tier-2 archive has DeBERTa-v3-base+CRF, not the recorded frozen DeBERTa-base+CRF	`reports/s179/optional_model_bundle_check_2026-11-12.json`

Strict-v2 is a separate explicit launcher, not a replacement benchmark. It reports a measured table point only with exact known declared food/process/target/state/basis/medium/time/temperature context and explicit numeric precision. It checks the final rendered fallback and refuses arbitrary generated text. No condition context is inferred by default, hence the zero retention-answer coverage in this sample. The source reports remain unverified and applicability is not certified. This is an engineering tradeoff, not evidence that answer quality improved.

Physical display metadata preserves TEXT_STATED, SCENARIO and MODEL_DERIVED distinctions and actual default/registry references. UNKNOWN remains UNKNOWN outside this sample. These references describe declared provenance, not scientific source truth. Recipe route chips retain legacy row-existence semantics and are not condition-applicability certificates. No new PARAM rows or valid real SIMULATE route are supplied. The earlier 30 frozen-answer flags, original H1/H3 failures, AA zero eligible SIM-VAL rows and zero T5 gate passes remain on record.

The wrong optional checkpoint was not substituted or run. Cross-family learned-NER versus Rung-A inference agreement remains data-dependent. Private 15a assignment rederivation, actual independent human labels, author contribution facts and future-dated acceptance remain outstanding. All work in this supplement ran locally without SSH or VPN.